

ELECTRONIC SKIN: Advancements And Opportunities



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*The various types of
artificial electronic skins
being developed offer
amazing properties and
applications*

*Ulsan National Institute's
electronic skin that can
detect changes in both
temperature and pressure*

Skin plays an important role in mediating our interactions with the world. Specifically, human skin can sense pressure and temperature, stretch, and heal itself. Electronic skin is a thin electronic material that mimics human skin in one or more ways. Recreating the properties of skin using electronic devices could have profound implications in various fields like robotics, prosthetics and medicine. The artificial skin could one day be used on robotic hands capable of detecting diseases or intoxication of humans via touch.

The pursuit of artificial skin has inspired innovations in materials to imitate skin's unique characteristics, including mechanical durability and stretch ability, biodegradability, and the ability to measure a diversity of complex sensations over large areas. New materials and fabrication strategies are being developed to make mechanically compliant and multifunctional skin-like electronics, and improve brain-machine interfaces that enable transmission of skin's signals into the body.

Scientists and engineers have made progress toward materials that can detect pressure, blend with surroundings, measure body temperature, and do much more. Here is a summary.

Self-healing electronic skin

Stanford University researchers have developed an electronic skin capable of healing itself by combining a self-healing plastic and nickel, a conductive metal. Unlike self-healing polymers developed by other researchers, this skin did not require a high temperature or UV light to activate.

The individual plastic molecules of the skin break apart relatively easily, but the bonds also easily reform. Cut pieces healed to 75 per cent strength within a few seconds and fully in less than 30 minutes when pressed together at room temperature. Additionally, the process could be repeated many times—in experiments the material showed near-perfect healing after 50 breaks. Other self-healing materials alter their structures in the process and thus can heal only once.

In addition to being self-healing, the electronic skin was pressure-sensitive and very flexible. It was the first material to exhibit all these properties at the same time. It was also the first conductive self-healing polymer. The e-skin could detect both downward pressure and pressure from bending; thus, in principle, it could detect both the pressure and angle of a normal human handshake.

According to the research team, the material could be useful in prosthetics and creating self-healing wires for electronic devices.

Lighting electronic skin

Researchers from the University of California at Berkeley have created an electronic skin that lights up when touched. Pressure triggered a reaction in the skin that lit up blue, green, red and yellow LEDs; as pressure increased the lights got brighter.

The material was composed of synthetic rubber and plastic, and was thinner than a piece of paper. Sandwiched between layers, organic LEDs were lit by semiconductor-enriched carbon nanotubes and a conductive silver ink. The skin was made up of hundreds of circuits, each of which con-

